PSTAT 126: REGRESSION ANALYSIS

Homework 2

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Section Time: Wednesday 5 PM

Question 1: Consider the simple linear regression model

$$Y = \beta_0 + \beta_1 x + \epsilon$$

where the intercept β_0 is known.

1. Find the least-squares estimator of β_1 for this model.
2. What is the variance of the slope $(\hat{\beta_1})$ for the least-squares estimator found in the above part?
   Consider the hypothesis test

$$H_0 : \beta_1 = 0 \text{ vs. } H_1 : \beta_1 \neq 0$$

3. What is the rejection region for the test?
4. What is the p-value of the test? (Bonus point)

1. $\min \sum_{i=1} \hat{\epsilon}_i^2 = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 = f \quad \frac{\partial f}{\partial \beta_1} = \sum_{i=1}^n 2(y_i - \beta_0 - \beta_1 x_i) x_i = 0$

$\sum_{i=1}^n y_i x_i - \beta_0 x_i - \beta_1 x_i \cdot x_i = 0 \quad \sum_{i=1}^n y_i x_i - \sum_{i=1}^n \beta_0 x_i - \sum_{i=1}^n \beta_1 x_i^2 = 0$

$\beta_1 \sum_{i=1}^n x_i^2 = \sum_{i=1}^n \beta_0 x_i - y_i x_i \quad \boxed{\hat{\beta_1} = \frac{\sum_{i=1}^n (\beta_0 - y_i) x_i}{\sum_{i=1}^n x_i^2}}$

2. $Var(\hat{\beta_1}) = Var\left(\frac{\sum_{i=1}^n (\beta_0 - y_i) x_i}{\sum_{i=1}^n x_i^2}\right) = \frac{1}{(\sum_{i=1}^n x_i^2)^2} Var\left(\sum_{i=1}^n x_i \beta_0 - y_i x_i\right)$

$= \frac{1}{(\sum_{i=1}^n x_i^2)^2}\left[Var\left(\sum_{i=1}^n x_i \beta_0\right) + Var\left(\sum_{i=1}^n y_i x_i\right) - Cov\left(\sum_{i=1}^n x_i \beta_0, \sum_{i=1}^n y_i x_i\right)\right]$

$\Rightarrow x_i$ & β_0 are known, so their variances are 0. Additionaly they are independent of y_i, so the covariance term is 0.

$= \frac{1}{(\sum_{i=1}^n x_i^2)^2}\left[\sum_{i=1}^n x_i^2 Var(y_i) + \cancel{2 \sum_{i \neq j} \sum Cov(y_i, y_j)}\right] \Rightarrow$ all y_i's are iid so $Cov(y_i, y_j) = 0$

$= \frac{1}{(\sum_{i=1}^n x_i^2)^2}\left[\sum_{i=1}^n x_i^2 \sigma^2\right] = \frac{\sigma^2 \cdot \sum_{i=1}^n x_i^2}{(\sum_{i=1}^n x_i^2)^2} = \boxed{\frac{\sigma^2}{\sum_{i=1}^n x_i^2}}$

3. The rejection region is anything outside $[-z_{\alpha/2}, z_{\alpha/2}]$ if σ^2 is known, or $[-t_{n-1,\alpha/2}, t_{n-1,\alpha/2}]$ if σ^2 is unknown. Use $n-1$ as the degrees of freedom because we are estimating the value for β_1.

4. $z = \frac{\hat{\beta_1}}{\sqrt{\frac{\sigma^2}{\sum_{i=1}^n x_i^2}}} \Rightarrow$ p-value $= 2 \cdot P(Z > |z|)$

$t = \frac{\hat{\beta_1}}{\sqrt{\frac{\hat{\sigma}^2}{\sum_{i=1}^n x_i^2}}} \Rightarrow$ p-value $= 2 \cdot P(T_{n-1} > |t|)$

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Question 2: Find the $(1-\alpha)$ confidence interval for the average value of y at the point $x = x_0$, when σ^2 is known and unknown using two-sided hypothesis testing.

$$E[\widehat{y|x_0}] = \hat{\beta}_0 + \hat{\beta}_1 x_0$$

$$Var[\widehat{y|x_0}] = \sigma^2\left[\frac{1}{n} + \frac{(x_0-\bar{x})^2}{\sum_{i=1}^{n}(x_i-\bar{x})^2}\right] , \quad \hat{\sigma}^2\left[\frac{1}{n} + \frac{(x_0-\bar{x})^2}{\sum_{i=1}^{n}(x_i-\bar{x})^2}\right]$$

if σ^2 known, if σ^2 unknown

if known use $Z_{\alpha/2}$ as critical value

if unknown use $t_{n-1,\alpha/2}$ as critical value

$$CI = (\hat{\beta}_0 + \hat{\beta}_1 x_0) \pm Z_{\alpha/2} \cdot \sigma\sqrt{\frac{1}{n} + \frac{(x_0-\bar{x})^2}{\sum_{i=1}^{n}(x_i-\bar{x})^2}} \quad \text{if } \sigma \text{ is known}$$

$$CI = (\hat{\beta}_0 + \hat{\beta}_1 x_0) \pm t_{n-2,\alpha/2} \cdot \hat{\sigma}\sqrt{\frac{1}{n} + \frac{(x_0-\bar{x})^2}{\sum_{i=1}^{n}(x_i-\bar{x})^2}} \quad \text{if } \sigma \text{ is unknown}$$

Question 3: Given the simple linear regression model:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

where $\epsilon_i \sim \mathcal{N}(0,1)$ and are independent and identically distributed, find the maximum likelihood estimates of $\beta_0, \beta_1, \sigma^2$.

$\epsilon_i \sim N(0,1) \Rightarrow y_i \sim (\hat{\beta}_0 + \hat{\beta}_1 x_i, 1)$ by the assumption

$$L(y_i, x_i; \beta_0, \beta_1, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{y_i-\mu}{\sigma}\right)^2\right) = \prod_{i=1}^{n} \left(\frac{1}{2\pi\sigma^2}\right)^{1/2} \exp\left(-\frac{(y_i-\beta_0-\beta_1 x_i)^2}{2\sigma^2}\right)$$

$$= \prod_{i=1}^{n} \left(\frac{1}{2\pi\sigma^2}\right)^{1/2} \cdot \prod_{i=1}^{n} \exp\left(-\frac{(y_i-\beta_0-\beta_1 x_i)^2}{2\sigma^2}\right) = \prod_{i=1}^{n} \left(\frac{1}{2\pi\sigma^2}\right)^{1/2} \cdot \exp\left(\sum_{i=1}^{n} \frac{-(y_i-\beta_0-\beta_1 x_i)^2}{2\sigma^2}\right)$$

$$= \prod_{i=1}^{n} \left(\frac{1}{2\pi\sigma^2}\right)^{1/2} \cdot \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i-\beta_0-\beta_1 x_i)^2\right)$$

plug in $\sigma^2 = 1 \Rightarrow \prod_{i=1}^{n} \left(\frac{1}{2\pi}\right)^{1/2} \cdot \exp\left(-\frac{1}{2}\sum_{i=1}^{n} (y_i-\beta_0-\beta_1 x_i)^2\right)$

$$l = \ln(L(y_i, x_i; \beta_0, \beta_1)) = \ln\left(\prod_{i=1}^{n} \left(\frac{1}{2\pi}\right)^{1/2}\right) + -\frac{1}{2}\sum_{i=1}^{n} (y_i-\beta_0-\beta_1 x_i)^2$$

$$= \sum_{i=1}^{n} \ln\left(\left(\frac{1}{2\pi}\right)^{1/2}\right) + \frac{1}{2}\sum_{i=1}^{n} (y_i-\beta_0-\beta_1 x_i)^2$$

$$= \sum_{i=1}^{n} \frac{1}{2}\ln\left(\frac{1}{2\pi}\right) + \frac{1}{2}\sum_{i=1}^{n} (y_i-\beta_0-\beta_1 x_i)^2$$

$$= -\frac{n}{2}\ln(2\pi) + \frac{1}{2}\sum_{i=1}^{n} (y_i-\beta_0-\beta_1 x_i)^2$$

$\frac{\partial l}{\partial \beta_0} = \sum_{i=1}^{n} (y_i-\beta_0-\beta_1 x_i) = 0 \Rightarrow$ same as $\hat{\beta}_0$

$$\boxed{\tilde{\beta}_0 = \hat{\beta}_0 = \bar{y} - \tilde{\beta}_1 \bar{x}}$$

$\frac{\partial l}{\partial \beta_1} = -\sum_{i=1}^{n} (y_i-\beta_0-\beta_1 x_i) x_i = 0 \Rightarrow$ same as $\hat{\beta}_1$

$$\boxed{\tilde{\beta}_1 = \frac{S_{xy}}{S_{xx}}}$$

$\hat{\sigma}^2 = 1$ by the consequence of the assumption that $\epsilon_i \sim N(0,1)$

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Question 4:

1. Prove that the maximum likelihood estimate of σ^2 is biased.
2. Prove that the maximum likelihood estimate of σ^2 is asymptotically unbiased.
3. Determine the bias in the maximum likelihood estimate of σ^2.
4. Prove that the variance of an estimate of σ^2 obtained using MLE method is less than the variance of an estimate of σ^2 obtained using OLS method.

1. $\tilde{\sigma}^2 = \frac{SSE}{n}$, since MLE is not model dependent.

$\hat{\sigma}^2 = \frac{SSE}{n-2} \Rightarrow SSE = \hat{\sigma}^2 (n-2)$

$\tilde{\sigma}^2 = \frac{(n-2)}{n} \hat{\sigma}^2 \Rightarrow E[\tilde{\sigma}^2] = E[\frac{(n-2)}{n} \hat{\sigma}^2] = \frac{n-2}{n} E[\hat{\sigma}^2] = \frac{(n-2)}{n} \sigma^2$

3. $Bias(\tilde{\sigma}^2) = \frac{(n-2)}{n} \sigma^2 - \frac{n}{n}\sigma^2 = -\frac{2}{n}\sigma^2 \neq \sigma^2$

$\Rightarrow \therefore \tilde{\sigma}^2$ is biased.

2. $\lim_{n \to \infty} E[\tilde{\sigma}^2] = \lim_{n \to \infty} \frac{n-2}{n} \sigma^2 = \sigma^2$ ✓

4. $Var(\tilde{\sigma}^2) = Var((\frac{n-2}{n}) \hat{\sigma}^2) = (\frac{n-2}{n})^2 Var(\hat{\sigma}^2)$

since n is positive and presumably > 2, $(\frac{n-2}{n})^2 < 1$.
this means $Var(\tilde{\sigma}^2)$ is a fraction of $Var(\hat{\sigma}^2)$, proving the variance of $\tilde{\sigma}^2$ is strictly less than $\hat{\sigma}^2$.

Question 5: Prove the following properties of β's.

1. Use the fact tht $\hat{\beta_1}$ and $\hat{\beta_0}$ are linear combination of y_i to find the distribution of $\hat{\beta_1}$ and $\hat{\beta_0}$.
2. $\frac{\hat{\beta_0}-\beta_0}{\sqrt{\sigma^2\left(\frac{1}{n}+\frac{\bar{x}^2}{s_{XX}}\right)}} \sim \mathcal{N}(0,1)$
3. $\frac{\hat{\beta_1}-\beta_1}{\sqrt{\frac{\sigma^2}{s_{XX}}}} \sim \mathcal{N}(0,1)$
4. $\frac{\hat{\beta_0}-\beta_0}{\sqrt{MSE\left(\frac{1}{n}+\frac{\bar{x}^2}{s_{XX}}\right)}} \sim t_{n-2}$
5. $\frac{\hat{\beta_1}-\beta_1}{\sqrt{\frac{MSE}{s_{XX}}}} \sim t_{n-2}$

1. $\hat{\beta_1} = \sum_{i=1} k_i y_i \Rightarrow$ the sum of n y_i's. Each $y_i \sim N(\hat{\beta_0} + \hat{\beta_1} x_i, \sigma^2)$ & the num of normal variables is a normal variable.

$\therefore \hat{\beta_1} \sim N(\beta_1, \frac{\sigma^2}{S_{xx}})$.

Similarly, since $\hat{\beta_0}$ is also a linear combination of y_i's, the same rule applies

$\Rightarrow \hat{\beta_0} \sim N(\beta_0, \sigma^2(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}))$

3. $\frac{\hat{\beta_1} - \beta_1}{\sqrt{\frac{\sigma^2}{S_{xx}}}} \Rightarrow \beta_1 = \mu_{\hat{\beta_1}} \Rightarrow \frac{\hat{\beta_1} - \mu_{\hat{\beta_1}}}{sd_{\hat{\beta_1}}} \Rightarrow$ standardizing a normal variable, so it is a standard normal variable.

$\Rightarrow \sqrt{\frac{\sigma^2}{S_{xx}}} = sd_{\hat{\beta_1}}$

$$\frac{\hat{\beta_1} - \beta_1}{\sqrt{\frac{\sigma^2}{S_{xx}}}} \sim Z = N(0,1)$$

4 $$\frac{\hat{\beta_1} - \beta_1}{\sqrt{MSE/S_{xx}}} \cdot \frac{\frac{1}{\sqrt{\sigma^2/S_{xx}}}}{\frac{1}{\sqrt{\sigma^2/S_{xx}}}} = \frac{\frac{\hat{\beta_1} - \beta_1}{\sqrt{\sigma^2/S_{xx}}}}{\sqrt{\frac{SSE/\sigma^2}{n-2}}} = \frac{Z}{\sqrt{\chi^2_{n-2}/(n-2)}} \sim t_{n-2}$$

5. $$\frac{\hat{\beta_0} - \beta_0}{\sqrt{MSE(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}})}} \cdot \frac{\frac{1}{\sqrt{\sigma^2(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}})}}}{\frac{1}{\sqrt{\sigma^2(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}})}}} = \frac{\frac{\hat{\beta_0} - \beta_0}{\sqrt{\sigma^2(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}})}}}{\sqrt{\frac{SSE/\sigma^2}{(n-2)}}} = \frac{Z}{\sqrt{\chi^2/(n-2)}} \sim t_{n-2}$$

2. $\frac{\hat{\beta_0} - \beta_0}{\sqrt{\sigma^2(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}})}} \Rightarrow \frac{\text{value} - \text{mean}}{sd} \Rightarrow$ standard normal

$$\Rightarrow \frac{\hat{\beta_0} - \beta_0}{\sqrt{\sigma^2(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}})}} \sim Z = N(0,1)$$

Question 6: Find the $(1-\alpha)$ prediction interval for y at the point $x = x_0$, when σ^2 is known and unknown using two-sided hypothesis testing.

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σ^2 is known $\Rightarrow$ use normal distribution

prediction interval $\rightarrow$ $Var(\widehat{y_i | x_0}) = Var(\hat{\beta}_0 + \hat{\beta}_1 x_0) + Var(\epsilon_i)$

$$= \sigma^2 + \sigma^2\left(\frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}\right)$$

$$= \sigma^2\left(1 + \frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}\right)$$

Critical value is $z_{\alpha/2}$

Prediction interval =

$$\hat{y}_0 \pm z_{\alpha/2} \cdot \sigma \sqrt{1 + \frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}}$$

$$= \left[\hat{y}_0 - z_{\alpha/2} \cdot \sigma \sqrt{1 + \frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}},\ \hat{y}_0 + z_{\alpha/2} \cdot \sigma \sqrt{1 + \frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}}\right]$$

When σ^2 is unknown:

$$Var(\widehat{y_i | x_0}) = MSE\left(1 + \frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}\right)$$

$\rightarrow$ use t distribution

critical value is $t_{n-2,\alpha/2}$

Prediction interval:

$$\hat{y}_0 \pm t_{n-2,\alpha/2} \cdot \hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}}$$

$$= \left[\hat{y}_0 - t_{n-2,\alpha/2} \cdot \hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}},\ \hat{y}_0 + t_{n-2,\alpha/2} \cdot \hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}}\right]$$

Question 7: Let $e_i := y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$, then prove that

1. $\sum_{i=1}^n e_i = 0$
2. $\sum_{i=1}^n x_i e_i = 0$
3. $\sum_{i=1}^n \hat{y}_i e_i = 0$
4. $\sum_{i=1}^n \hat{y}_i = \sum_{i=1}^n y_i$
5. The least-squares regression line always passes through the centroid $(\bar{x}, \bar{y})$ of the data.

1. $\sum_{i=1}^n e_i = \sum_{i=1}^n y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i = \sum_{i=1}^n y_i - \bar{y} + \hat{\beta}_1 \bar{x} - \hat{\beta}_1 x_i = \sum_{i=1}^n (y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x})$

$= \sum_{i=1}^n y_i - \bar{y} - \hat{\beta}_1 \sum_{i=1}^n x_i - \bar{x} = n\bar{y} - n\bar{y} - \hat{\beta}_1 (n\bar{x} - n\bar{x}) = 0 - \hat{\beta}_1 \cdot 0$

$= 0$ ✓

2. $\sum_{i=1}^n x_i e_i = \sum_{i=1}^n x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = \sum_{i=1}^n x_i y_i - \hat{\beta}_0 x_i - \hat{\beta}_1 x_i$

$= \sum_{i=1}^n x_i y_i - x_i \bar{y} + \hat{\beta}_1 \bar{x} x_i - \hat{\beta}_1 x_i^2 = \sum_{i=1}^n x_i (y_i - \bar{y}) - \hat{\beta}_1 \sum_{i=1}^n x_i (x_i - \bar{x})$

$= S_{xy} - \hat{\beta}_1 S_{xx}$ $\qquad \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$

$= S_{xy} - \frac{S_{xy}}{\cancel{S_{xx}}} \cdot \cancel{S_{xx}} = S_{xy} - S_{xy} = 0$ ✓

3. $\sum_{i=1}^n \hat{y}_i e_i = 0 = \sum_{i=1}^n (\hat{\beta}_0 + \hat{\beta}_1 x_i) e_i = \sum_{i=1}^n \hat{\beta}_0 e_i + \sum_{i=1}^n \hat{\beta}_1 x_i e_i$

$= \hat{\beta}_0 \sum_{i=1}^n e_i + \hat{\beta}_1 \sum_{i=1}^n x_i e_i = \hat{\beta}_0 \cdot 0 + \hat{\beta}_1 \cdot 0 = 0$ ✓

4. $e_i = y_i - \hat{y}_i \Rightarrow y_i = \hat{y}_i + e_i$

$\sum_{i=1}^n y_i = \sum_{i=1}^n \hat{y}_i + e_i = \sum_{i=1}^n \hat{y}_i + \cancel{\sum_{i=1}^n e_i} = \sum_{i=1}^n \hat{y}_i$ ✓

5. $\sum_{i=1}^n e_i = 0 \Rightarrow \sum_{i=1}^n y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i = 0 \rightarrow n\bar{y} - n\hat{\beta}_0 - n\bar{x}\hat{\beta}_1 = 0$

$\Rightarrow \bar{y} - \hat{\beta}_0 - \bar{x}\hat{\beta}_1 = 0 \rightarrow \bar{y} = \hat{\beta}_0 + \bar{x}\hat{\beta}_1$

$\Rightarrow$ shows that $(\bar{x}, \bar{y})$ is a point on the graph, regardless of what n, $\hat{\beta}_0$, + $\hat{\beta}_1$ are.

CODING

1. Using the same dataset as stated in Homework 1, create a null model and a simple linear regression model. Plot the regression line passing through the points. Explain all the elements of the output obtained after calling the summary function.

2. Find the 95% confidence interval for all estimates and the 95% prediction interval.

3. Based on various measures such as residual standard error, coefficient of determination and F-value, comment on which model you would prefer.

4. Finally, interpret all the possible measures in the output, including the estimates, their standard errors, t-value, p-value, residual standard error, R^2, F value.

In Rmd

hw2

Mitul Marimuthu

2026-05-10

```r
data <- read.csv("./datasets/Dataset.csv")
```

```r
class(data['Energy_Consumption_kWh'])
head(data['Energy_Consumption_kWh'])
```

```r
null_model <- lm(Energy_Consumption_kWh ~ 1, data = data)
```

```r
null_model$coefficients
```

```
## (Intercept) 
##    102.8555
```

```r
mean(data$Energy_Consumption_kWh)
```

```
## [1] 102.8555
```

```r
sse_null <- sum(resid(null_model)^2)
sse_null
```

```
## [1] 16436.42
```

```r
summary(null_model)
```

```
## 
## Call:
## lm(formula = Energy_Consumption_kWh ~ 1, data = data)
## 
## Residuals:
##     Min      1Q  Median      3Q     Max 
## -28.178 -14.780   0.021  11.335  39.273 
## 
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)    
## (Intercept)   102.86       2.59   39.71   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 
## Residual standard error: 18.31 on 49 degrees of freedom
```

```
sim_lin_reg = lm(data$Energy_Consumption_kWh ~ data$Temperature_C, data = data)
```

```
summary(sim_lin_reg)
```

```
##
## Call:
## lm(formula = data$Energy_Consumption_kWh ~ data$Temperature_C,
##     data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.5034  -2.4541   0.0241   3.4235   8.3998
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)        50.93031    2.04439   24.91   <2e-16 ***
## data$Temperature_C  2.45531    0.09158   26.81   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.63 on 48 degrees of freedom
## Multiple R-squared:  0.9374, Adjusted R-squared:  0.9361
## F-statistic: 718.8 on 1 and 48 DF,  p-value: < 2.2e-16
```

```
sse_lin <- sum(resid(sim_lin_reg)^2)
```

```
sse_null
```

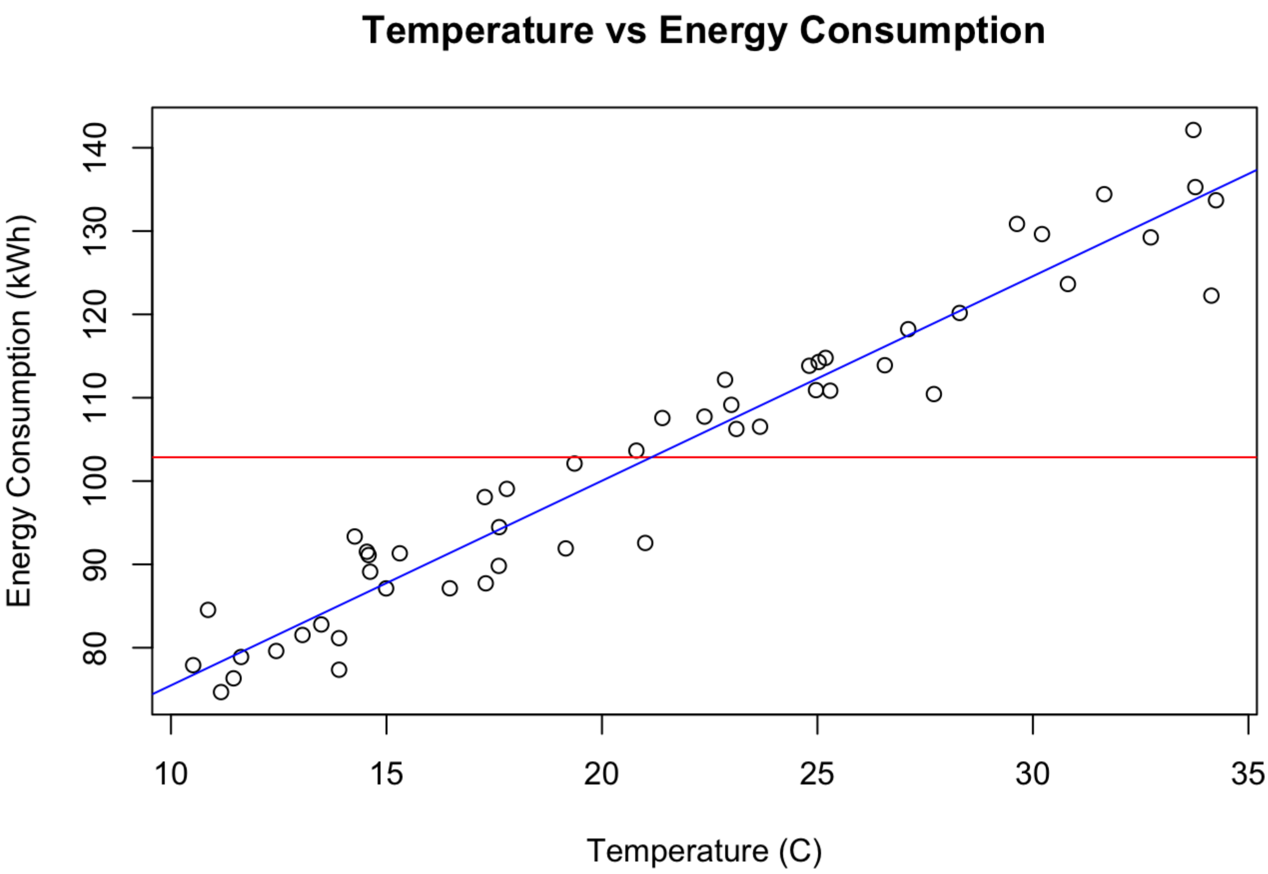
```
## [1] 16436.42
```

```
sse_lin
```

```
## [1] 1028.839
```

The simple linear regression model has a smaller SSE, therefore it is a better model.

```
plot(data$Temperature_C, data$Energy_Consumption_kWh, xlab = "Temperature (C)",
     ylab = "Energy Consumption (kWh)", main = "Temperature vs Energy Consumption")
abline(null_model, col = "red")
abline(sim_lin_reg, col = "blue")
```




```
summary(null_model)
```

```
##
## Call:
## lm(formula = Energy_Consumption_kWh ~ 1, data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -28.178 -14.780   0.021  11.335  39.273
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   102.86       2.59   39.71   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 18.31 on 49 degrees of freedom
```

1a. Null Model summary description

The residuals section shows the distribution of the residuals with min at -28.178 and max at 39.273.

Coefficents section:

- The value for $\hat{\beta}_0$ is 102.86, which is the mean of the data in the null model

- The standard error is 2.59, indicating that estimates for $\hat{\beta}_0$ deviate from the population parameter by 2.59 on average.
- The t-value is 39.71 meaning the value for $\hat{\beta}_0$ is 39.71 standard errors away from 0, and it's corresponding p-value (with a null hypothesis of $\beta_0 = 0$) is very small. This indicates that there is statistically significant evidence that $\beta_0 \neq 0$.
- A residual standard error of 18.31 indicates that the actual values deviate from their predicted values by 18.31 on average.

```
summary(sim_lin_reg)
```

```
##
## Call:
## lm(formula = data$Energy_Consumption_kWh ~ data$Temperature_C,
##     data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.5034  -2.4541   0.0241   3.4235   8.3998
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)         50.93031    2.04439   24.91   <2e-16 ***
## data$Temperature_C  2.45531    0.09158   26.81   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.63 on 48 degrees of freedom
## Multiple R-squared:  0.9374, Adjusted R-squared:  0.9361
## F-statistic: 718.8 on 1 and 48 DF,  p-value: < 2.2e-16
```

1b. Simple Linear Regression Model summary description

The residuals section shows the distribution of the residuals, with a min of -12.5034 and a max od 8.3998

Coefficients section:

- Intercept: $\hat{\beta}_0$ = 50.93031. It's standard error is 2.04439 meaning estimates for the intercept deviate from the true population parameter by 2.04439. It's t-value ie 24.91 meaning $\hat{\beta}_0$ is 24.91 stanard errors away from 0 which produces a p-value that is very small, indicating that there is strong statistical evidence that $\hat{\beta}_0 \neq 0$.
- `data$Temperature_C`: $\hat{\beta}_1$ = 2.45531. It's standard error is 0.09158 meaning estimates for the slope deviate from the true population parameter by 0.09158 on average. It's t-value is 26.81 meaning $\hat{\beta}_1$ is 26.81 standard errors away from 0 which produces a p-value that is very small, indicating that there is strong statistical evidence that $\hat{\beta}_1 \neq 0$.
- The residual standard error is 4.63, indicating the average deviation of 4.63 of actual values from their predicted values

- R^2 = 0.9374 indicating that around 93.74% of the variance in energy consumption is explained by temperature
- A large F-statistic and low p-value indicate that this model is better than just using the mean.

2. 95% confidence intervals and prediction intervals.

Null-model confidence interval for all estimates

```
ci = predict(null_model, interval="confidence", level = 0.95)
ci
```

```
##         fit      lwr      upr
## 1  102.8555 97.65047 108.0606
## 2  102.8555 97.65047 108.0606
## 3  102.8555 97.65047 108.0606
## 4  102.8555 97.65047 108.0606
## 5  102.8555 97.65047 108.0606
## 6  102.8555 97.65047 108.0606
## 7  102.8555 97.65047 108.0606
## 8  102.8555 97.65047 108.0606
## 9  102.8555 97.65047 108.0606
## 10 102.8555 97.65047 108.0606
## 11 102.8555 97.65047 108.0606
## 12 102.8555 97.65047 108.0606
## 13 102.8555 97.65047 108.0606
## 14 102.8555 97.65047 108.0606
## 15 102.8555 97.65047 108.0606
## 16 102.8555 97.65047 108.0606
## 17 102.8555 97.65047 108.0606
## 18 102.8555 97.65047 108.0606
## 19 102.8555 97.65047 108.0606
## 20 102.8555 97.65047 108.0606
## 21 102.8555 97.65047 108.0606
## 22 102.8555 97.65047 108.0606
## 23 102.8555 97.65047 108.0606
## 24 102.8555 97.65047 108.0606
## 25 102.8555 97.65047 108.0606
## 26 102.8555 97.65047 108.0606
## 27 102.8555 97.65047 108.0606
## 28 102.8555 97.65047 108.0606
## 29 102.8555 97.65047 108.0606
## 30 102.8555 97.65047 108.0606
## 31 102.8555 97.65047 108.0606
## 32 102.8555 97.65047 108.0606
## 33 102.8555 97.65047 108.0606
## 34 102.8555 97.65047 108.0606
## 35 102.8555 97.65047 108.0606
## 36 102.8555 97.65047 108.0606
## 37 102.8555 97.65047 108.0606
## 38 102.8555 97.65047 108.0606
## 39 102.8555 97.65047 108.0606
```

```
## 40 102.8555 97.65047 108.0606
## 41 102.8555 97.65047 108.0606
## 42 102.8555 97.65047 108.0606
## 43 102.8555 97.65047 108.0606
## 44 102.8555 97.65047 108.0606
## 45 102.8555 97.65047 108.0606
## 46 102.8555 97.65047 108.0606
## 47 102.8555 97.65047 108.0606
## 48 102.8555 97.65047 108.0606
## 49 102.8555 97.65047 108.0606
## 50 102.8555 97.65047 108.0606
```

Null-model prediction interval

```r
pi = predict(null_model, interval="prediction", level = 0.95)
```

```
## Warning in predict.lm(null_model, interval = "prediction", level = 0.95): prediction
s on current data refer to _future_ responses
```

```r
pi
```

```
##         fit      lwr     upr
## 1  102.8555 65.68403 140.027
## 2  102.8555 65.68403 140.027
## 3  102.8555 65.68403 140.027
## 4  102.8555 65.68403 140.027
## 5  102.8555 65.68403 140.027
## 6  102.8555 65.68403 140.027
## 7  102.8555 65.68403 140.027
## 8  102.8555 65.68403 140.027
## 9  102.8555 65.68403 140.027
## 10 102.8555 65.68403 140.027
## 11 102.8555 65.68403 140.027
## 12 102.8555 65.68403 140.027
## 13 102.8555 65.68403 140.027
## 14 102.8555 65.68403 140.027
## 15 102.8555 65.68403 140.027
## 16 102.8555 65.68403 140.027
## 17 102.8555 65.68403 140.027
## 18 102.8555 65.68403 140.027
## 19 102.8555 65.68403 140.027
## 20 102.8555 65.68403 140.027
## 21 102.8555 65.68403 140.027
## 22 102.8555 65.68403 140.027
## 23 102.8555 65.68403 140.027
## 24 102.8555 65.68403 140.027
## 25 102.8555 65.68403 140.027
## 26 102.8555 65.68403 140.027
## 27 102.8555 65.68403 140.027
## 28 102.8555 65.68403 140.027
## 29 102.8555 65.68403 140.027
```

```
## 30 102.8555 65.68403 140.027
## 31 102.8555 65.68403 140.027
## 32 102.8555 65.68403 140.027
## 33 102.8555 65.68403 140.027
## 34 102.8555 65.68403 140.027
## 35 102.8555 65.68403 140.027
## 36 102.8555 65.68403 140.027
## 37 102.8555 65.68403 140.027
## 38 102.8555 65.68403 140.027
## 39 102.8555 65.68403 140.027
## 40 102.8555 65.68403 140.027
## 41 102.8555 65.68403 140.027
## 42 102.8555 65.68403 140.027
## 43 102.8555 65.68403 140.027
## 44 102.8555 65.68403 140.027
## 45 102.8555 65.68403 140.027
## 46 102.8555 65.68403 140.027
## 47 102.8555 65.68403 140.027
## 48 102.8555 65.68403 140.027
## 49 102.8555 65.68403 140.027
## 50 102.8555 65.68403 140.027
```

Simple linear regression model confidence interval

```
ci_slr = predict(sim_lin_reg, interval="confidence", level=0.95)
ci_slr
```

```
##          fit       lwr       upr
## 1   98.47378  97.11695  99.83061
## 2  133.84098 131.17030 136.51166
## 3  120.41531 118.55328 122.27733
## 4  112.23080 110.73837 113.72322
## 5   85.06031  83.18576  86.93486
## 6   85.05883  83.18420  86.93346
## 7   79.04878  76.83058  81.26699
## 8  128.65179 126.31183 130.99175
## 9  112.38159 110.88380 113.87937
## 10 118.94695 117.16110 120.73279
## 11  76.74698  74.38762  79.10634
## 12 135.01926 132.27135 137.76716
## 13 126.58113 124.36782 128.79444
## 14  88.51742  86.81766  90.21718
## 15  86.64438  84.85246  88.43630
## 16  86.74133  84.95434  88.52833
## 17  94.15870  92.68955  95.62784
## 18 107.69448 106.32894 109.06002
## 19 101.99745 100.67944 103.31547
## 20  93.35992  91.86322  94.85661
## 21 113.04071 111.51873 114.56269
## 22  84.04597  82.11653  85.97542
## 23  93.41611  91.92141  94.91081
## 24  97.97177  96.60534  99.33821
```

```
## 25 103.47831 102.16104 104.79558
## 26 123.67977 121.63727 125.72227
## 27  87.73999  86.00276  89.47722
## 28 107.04861 105.69514 108.40209
## 29 111.84753 110.36843 113.32663
## 30  78.33470  76.07318  80.59623
## 31 112.77627 111.26414 114.28840
## 32  85.95070  84.12308  87.77831
## 33  79.47650  77.28403  81.66897
## 34 133.72873 131.06537 136.39208
## 35 134.75667 132.02604 137.48731
## 36 125.10516 122.97981 127.23051
## 37  94.18150  92.71312  95.64989
## 38  81.47884  79.40449  83.55319
## 39 117.48361 115.77001 119.19721
## 40 102.50125 101.18454 103.81796
## 41  82.97450  80.98556  84.96344
## 42 105.87881 104.54298 107.21463
## 43  77.59431  75.28741  79.90121
## 44 131.30011 128.79346 133.80675
## 45  91.36809  89.79483  92.94136
## 46 116.15094 114.49953 117.80235
## 47  94.61715  93.16295  96.07136
## 48 107.40669 106.04673 108.76666
## 49 109.04207 107.64627 110.43787
## 50  86.83034  85.04785  88.61283
```

Simple linear regression model prediction interval

```
pi_slr = predict(sim_lin_reg, interval="prediction", level=0.95)
```

```
## Warning in predict.lm(sim_lin_reg, interval = "prediction", level = 0.95): predictio
ns on current data refer to _future_ responses
```

```
pi_slr
```

```
##          fit       lwr       upr
## 1   98.47378  89.06677 107.88079
## 2  133.84098 124.15680 143.52516
## 3  120.41531 110.92226 129.90835
## 4  112.23080 102.80328 121.65832
## 5   85.06031  75.56480  94.55582
## 6   85.05883  75.56331  94.55436
## 7   79.04878  69.47950  88.61807
## 8  128.65179 119.05355 138.25003
## 9  112.38159 102.95322 121.80996
## 10 118.94695 109.46855 128.42534
## 11  76.74698  67.14399  86.34997
## 12 135.01926 125.31350 144.72502
## 13 126.58113 117.01298 136.14928
## 14  88.51742  79.05487  97.97998
```

```
## 15  86.64438  77.16483  96.12392
## 16  86.74133  77.26272  96.21995
## 17  94.15870  84.73483 103.58256
## 18 107.69448  98.28621 117.10275
## 19 101.99745  92.59597 111.39894
## 20  93.35992  83.93172 102.78811
## 21 113.04071 103.60847 122.47295
## 22  84.04597  74.53947  93.55247
## 23  93.41611  83.98823 102.84399
## 24  97.97177  88.56338 107.38017
## 25 103.47831  94.07693 112.87969
## 26 123.67977 114.14968 133.20986
## 27  87.73999  78.27063  97.20935
## 28 107.04861  97.64209 116.45514
## 29 111.84753 102.42211 121.27295
## 30  78.33470  68.75528  87.91412
## 31 112.77627 103.34561 122.20693
## 32  85.95070  76.46434  95.43706
## 33  79.47650  69.91314  89.03985
## 34 133.72873 124.04656 143.41089
## 35 134.75667 125.05579 144.45756
## 36 125.10516 115.55697 134.65335
## 37  94.18150  84.75776 103.60524
## 38  81.47884  71.94187  91.01580
## 39 117.48361 108.01855 126.94866
## 40 102.50125  93.09995 111.90256
## 41  82.97450  73.45574  92.49325
## 42 105.87881  96.47481 115.28281
## 43  77.59431  68.00408  87.18454
## 44 131.30011 121.65987 140.94034
## 45  91.36809  81.92744 100.80875
## 46 116.15094 106.69695 125.60493
## 47  94.61715  85.19561 104.03870
## 48 107.40669  97.99923 116.81415
## 49 109.04207  99.62936 118.45478
## 50  86.83034  77.35257  96.30810
```

3. Which model

The simple linear regression model has a smaller RSE, higher R^2 value and higher F-value so therefore I would rather have the simple linear regression model over the null model.

4. Interpretations

4a. Null Model

- The intercept is 102.86, meaning that the energy consumed is 102.86 kWh on average.
- The standard error is 2.59, indicating that estimates for the intercept deviate from the population parameter by 2.59 on average.

- The t-value is 39.71 meaning the value for $\hat{\beta}_0$ is 39.71 standard errors away from 0, and it's corresponding p-value (with a null hypothesis of $\beta_0 = 0$) is very small. This indicates that there is statistically significant evidence that $\beta_0 \neq 0$.

4b. Simple Linear Regression Model

- Intercept: $\hat{\beta}_0$ = 50.93031. This means that when the temperature is 0 degrees Celsius, the energy consumed in 50.93031 kWh. It's standard error is 2.04439 meaning estimates for the intercept deviate from the true population parameter by 2.04439. It's t-value ie 24.91 meaning $\hat{\beta}_0$ is 24.91 stanard errors away from 0 which produces a p-value that is very small, indicating that there is strong statistical evidence that $\hat{\beta}_0 \neq 0$.
- `data$Temperature_C`: $\hat{\beta}_1$ = 2.45531. This means that for each 1 degree Celsius increase in temperature, the energy consumed increases by 2.45531 kWh. It's standard error is 0.09158 meaning estimates for the slope deviate from the true population parameter by 0.09158 on average. It's t-value is 26.81 meaning $\hat{\beta}_1$ is 26.81 standard errors away from 0 which produces a p-value that is very small, indicating that there is strong statistical evidence that $\hat{\beta}_1 \neq 0$.
- The residual standard error is 4.63, indicating the average deviation of 4.63 of actual values from their predicted values
- R^2 = 0.9374 indicating that around 93.74% of the variance in energy consumption is explained by temperature
- A large F-statistic and low p-value indicate that this model is better than just using the mean.